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探究 CTRL+C 和 CTRL+V 对大学生写作业和论文的帮助

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摘要

The ubiquitous keyboard shortcuts CTRL+C (copy) and CTRL+V (paste) have fundamentally transformed academic writing practices among university students. While these digital tools enhance information processing efficiency and facilitate literature review compilation, their misuse has precipitated an epidemic of plagiarism and academic dishonesty in higher education. This study investigates the multifaceted impacts of copy-paste functionality on students' assignment completion and thesis development, drawing upon recent empirical data from 2024-2025 institutional surveys involving over 3,000 faculty members and students across Chinese universities. Our analysis reveals that nearly 60% of university faculty and students utilize generative AI tools daily or weekly, with approximately 30% of students primarily employing these technologies for essay writing and assignment completion. Furthermore, the evolution from simple text duplication to AI-generated content replication has exacerbated academic integrity challenges, prompting stringent institutional responses including zero-tolerance policies and AI detection implementation. This paper examines the technological affordances of digital copying, psychological mechanisms underlying student reliance on these shortcuts, institutional detection and prevention strategies, and pedagogical recommendations for fostering authentic academic literacy in the digital age.

Keywords: Academic integrity; plagiarism; digital literacy; copy-paste behavior; AI-generated content; higher education

键盘快捷键 CTRL+C（复制）和 CTRL+V（粘贴）的普及从根本上改变了大学生的学术写作实践。虽然这些数字工具提高了信息处理效率并促进了文献综述的编制，但其滥用已在高等教育中引发了抄袭和学术不端的流行。本研究基于 2024-2025 年对中国高校 3000 多名师生进行的机构调查实证数据，调查了复制粘贴功能对学生作业完成和论文发展的多方面影响。分析显示，近 60%的高校师生每天或每周使用生成式 AI 工具，约 30%的学生主要将这些技术用于论文写作和作业完成。此外，从简单文本复制到 AI 生成内容复制的演变加剧了学术诚信挑战，促使机构采取零容忍政策和 AI 检测实施等严格应对措施。本文探讨了数字复制的技术可供性、学生依赖这些快捷键的心理机制、机构检测和预防策略，以及在数字时代培养真实学术素养的教学建议。

关键词：学术诚信；抄袭；数字素养；复制粘贴行为；AI 生成内容；高等教育

Introduction 引言

1.1 The Digital Revolution in Academic Writing / 学术写作中的数字革命

The advent of personal computing and word processing software in the late 20th century introduced unprecedented efficiency to academic writing. Among the most transformative innovations were the keyboard shortcuts CTRL+C and CTRL+V, which enabled instantaneous duplication of digital text. Originally designed to streamline document editing and reduce repetitive typing, these commands have become deeply embedded in the cognitive workflows of contemporary students. However, the same functionality that facilitates legitimate research synthesis has also enabled systematic intellectual appropriation.

20 世纪末个人计算机和文字处理软件的出现为学术写作带来了前所未有的效率。其中最具变革性的创新是键盘快捷键

CTRL+C 和 CTRL+V，它们实现了数字文本的即时复制。这些命令最初旨在简化文档编辑和减少重复输入，现已成为当代学生认知工作流程中根深蒂固的组成部分。然而，这一促进合法研究综合的功能也促成了系统性的知识挪用。

Recent investigations indicate that the copy-paste paradigm has evolved beyond simple text duplication. A 2025 survey by MyCos (麦可思), a third-party educational research institution, revealed that generative AI tools have become "essential artifacts" for university students completing reports and essays. The study of over 3,000 faculty members and students found that nearly 60% use generative AI daily or weekly, with approximately 30% of university students primarily deploying these tools for thesis writing and assignment completion. Alarmingly, interview data confirmed that some students directly copy and paste AI-generated content into their submissions without substantive modification or critical evaluation.

最近的研究表明，复制粘贴范式已超越简单的文本复制。第三方教育研究机构麦可思 2025 年的一项调查显示，生成式 AI 工具已成为大学生完成报告和论文的"必备神器"。这项针对 3000 多名师生的研究发现，近 60%的人每天或每周使用生成式 AI，约 30%的大学生主要将这些工具用于论文写作和作业完成。令人担忧的是，访谈数据证实，一些学生在没有实质性修改或批判性评估的情况下，直接将 AI 生成的内容复制粘贴到提交的作业中。

1.2 Statement of the Problem / 问题陈述

The normalization of copy-paste behaviors in academic contexts presents a complex pedagogical challenge. While information digitization has democratized access to knowledge, it has simultaneously blurred the boundaries between legitimate research appropriation and academic misconduct. The ease with which students can duplicate content—from peer-reviewed articles to AI-generated paragraphs—has outpaced the development of corresponding ethical frameworks and detection methodologies.

学术环境中复制粘贴行为的常态化提出了复杂的教学挑战。虽然信息数字化使知识获取民主化，但它同时模糊了合法研究挪用与学术不端之间的界限。学生可以轻松复制内容——从同行评审文章到 AI 生成段落——这种便利性超越了相应伦理框架和检测方法的发展速度。

Contemporary academic integrity policies struggle to address the nuanced spectrum of copy-paste behaviors. These range from technically permissible but educationally problematic practices (such as excessive quotation without synthesis) to explicitly fraudulent activities (including data fabrication through AI image editing and wholesale submission of machine-generated theses). The investigation team from Zhejiang University's College of Education, led by Huang Yating, conducted three waves of surveys beginning in early 2024, encompassing over 3,800 undergraduates and 4,200 graduate students across more than 20 Chinese universities. Their findings indicate that inappropriate use of generative AI is prevalent among students, with significant proportions reporting habitual reliance on AI for task completion.

当代学术诚信政策难以应对复制粘贴行为的细微频谱。这些行为范围从技术允许但教育上有问题的做法（如缺乏综合的过度引用）到明确欺诈的活动（包括通过 AI 图像编辑伪造数据和整篇提交机器生成的论文）。浙江大学教育学院黄亚婷团队从 2024 年初开始进行了三波调查，涵盖 20 多所中国高校的 3800 多名本科生和 4200 多名研究生。他们的发现表明，生成式 AI 的不当使用在学生中普遍存在，相当比例的学生报告习惯依赖 AI 完成任务。

1.3 Research Objectives / 研究目标

This study aims to: (1) characterize the technological and psychological mechanisms underlying student reliance on CTRL+C/CTRL+V functionalities; (2) analyze the prevalence and manifestations of copy-paste behaviors in contemporary university settings; (3) evaluate institutional responses including detection technologies and policy frameworks; and (4) propose evidence-based pedagogical interventions to promote authentic academic writing.

本研究旨在：(1)描述学生依赖 CTRL+C/CTRL+V 功能的技术和心理机制；(2)分析当代大学环境中复制粘贴行为的普遍性和表现形式；(3)评估机构应对措施，包括检测技术和政策框架；(4)提出基于证据的教学干预措施，以促进真实的学术写作。

Technological Affordances and Cognitive Impact / 技术可供性与认知影响

2.1 The Architecture of Digital Reproduction / 数字复制的架构

The CTRL+C and CTRL+V commands function through the operating system's clipboard mechanism, a temporary storage buffer that enables data transfer between applications. This architectural feature, while technologically neutral, possesses inherent affordances that shape user behavior. The frictionless nature of digital copying—requiring minimal physical effort and cognitive load—contrasts sharply with traditional note-taking practices that necessitate manual transcription and thereby encourage deeper cognitive processing.

CTRL+C 和 CTRL+V 命令通过操作系统的剪贴板机制运行，这是一个临时存储缓冲区，可实现应用程序之间的数据传输。这一架构特征虽然在技术上是中性的，但具有塑造用户行为的内在可供性。数字复制的无摩擦特性——需要最小的体力努力和认知负荷——与需要手动转录并从而促进更深层次认知处理的传统笔记实践形成鲜明对比。

Research in educational psychology suggests that the "copy-paste effect" may impair knowledge retention and conceptual understanding. When students manually transcribe information, they engage in active processing that facilitates encoding into long-term memory. Conversely, the mechanical duplication enabled by keyboard shortcuts bypasses these cognitive elaboration stages, potentially resulting in superficial engagement with source material.

教育心理学研究表明，"复制粘贴效应"可能损害知识保持和概念理解。当学生手动转录信息时，他们参与促进编码到长期记忆的主动处理。相反，键盘快捷键实现的机械复制绕过了这些认知 elaboration 阶段，可能导致对源材料的表面参与。

2.2 From Textual Duplication to AI-Assisted Composition / 从文本复制到 AI 辅助写作

The copy-paste paradigm has undergone significant transformation with the proliferation of large language models

(LLMs). Contemporary students now engage in "second-order copying"—duplicating content that was itself algorithmically generated rather than human-authored. This shift introduces novel epistemological challenges, as AI-generated text often exhibits grammatical correctness and structural coherence while potentially containing factual inaccuracies or logical fallacies.
随着大型语言模型(LLM)的普及，复制粘贴范式经历了重大转变。当代学生现在从事"二阶复制"——复制本身由算法生成而非人类创作的内容。这种转变引入了新的认识论挑战，因为 AI 生成的文本通常表现出语法正确性和结构连贯性，同时可能包含事实不准确或逻辑谬误。
According to Ding Junpeng, research assistant at the Engineering Research Center of the Ministry of Education, the most egregious violations involve using AI tools to automatically generate entire papers. More commonly, students utilize generative AI to fabricate research data, edit experimental images, and substitute autonomous design processes. These practices represent an escalation from traditional plagiarism, as AI technologies significantly reduce the cost and complexity of academic fraud.
据教育部信息网络工程研究中心科研助理丁峻鹏介绍，最严重的违规行为涉及使用 AI 工具自动生成整篇论文。更常见的是，学生利用生成式 AI 伪造研究数据、编辑实验图像并替代自主设计过程。这些做法代表了从传统抄袭的升级，因为 AI 技术显著降低了学术欺诈的成本和复杂性。

2.3 Cognitive Offloading and Skill Atrophy / 认知卸载与技能萎缩

Neuroscientific research on "cognitive offloading" provides insight into the long-term consequences of habitual copy-paste reliance. When students consistently outsource composition tasks to digital tools, they may experience atrophy in critical academic skills including synthesis, paraphrasing, and independent argumentation. Student interviews conducted in recent media investigations reveal concerning patterns: "AI can quickly generate content for me, saving a large chunk of the thinking process. When thinking independently, I experience some difficulty aversion." This testimony illustrates the psychological dependency that can develop when technological shortcuts substitute for cognitive effort.
关于"认知卸载"的神经科学研究为习惯性依赖复制粘贴的长期后果提供了洞见。当学生持续将写作任务外包给数字工具时，他们可能会经历包括综合、改写和独立论证在内的关键学术技能的萎缩。最近媒体调查进行的学生访谈揭示了令人担忧的模式："AI 可以很快帮我生成，一大块思考过程就省下来了。自主思考时，就会有点畏难情绪。"这一证词说明了当技术捷径替代认知努力时可能发展出的心理依赖。
The phenomenon of "AI dependency" documented by Zhejiang University researchers suggests that prolonged reliance on copy-paste functionalities—whether for human-authored or AI-generated content—may fundamentally alter students' approaches to knowledge production. Students reporting habitual AI assistance demonstrated reduced confidence in independent problem-solving and increased anxiety when confronted with open-ended analytical tasks.
浙江大学研究人员记录的"AI 依赖"现象表明，长期依赖复制粘贴功能——无论是用于人类创作还是 AI 生成的内容——可能从根本上改变学生的知识生产方法。报告习惯性 AI 辅助的学生表现出对独立解决问题信心的降低，以及在面对开放式分析任务时焦虑的增加。

3 Prevalence and Manifestations of Copy-Paste Behaviors / 复制粘贴行为的普遍性与表现形式

3.1 Quantitative Patterns of Misconduct / 不端行为的定量模式

Empirical data from multiple institutional surveys conducted between 2024 and 2025 reveal alarming prevalence rates for copy-paste-related academic misconduct. The MyCos survey indicates that among students utilizing generative AI, nearly 30% primarily employ these tools for thesis writing and assignment completion. This statistic suggests that a substantial minority of students have integrated digital duplication into their standard academic workflow.
2024 至 2025 年进行的多项机构调查实证数据揭示了与复制粘贴相关的学术不端的惊人普遍率。麦可思调查显示，在使用生成式 AI 的学生中，近 30%主要将这些工具用于论文写作和作业完成。这一统计表明，相当比例的学生已将数字复制整合到其标准学术工作流程中。
Historical comparisons demonstrate the escalating nature of this phenomenon. Research published in the *International Journal of Multidisciplinary: Applied Business and Education Research* documented that plagiarism among pre-service teachers often manifests through "CTRL C + CTRL V" behaviors, particularly regarding citation and referencing knowledge deficits. More recent data suggest that the transition from traditional copy-paste to AI-assisted duplication has accelerated these trends, with some instructors reporting that AI-generated submissions now constitute the majority of academic integrity violations in writing-intensive courses.
历史比较表明了这一现象的升级性质。发表在《国际多学科：应用商业与教育研究期刊》上的研究记录了职前教师中的抄袭行为通常通过"CTRL C + CTRL V"行为表现出来，特别是在引用和参考文献知识缺陷方面。更新的数据表明，从传统复制粘贴到 AI 辅助复制的转变加速了这些趋势，一些教师报告 AI 生成的提交现在构成了写作密集型课程中学术诚信违规的大部分。

3.2 Typology of Copy-Paste Violations / 复制粘贴违规类型学

Contemporary academic misconduct involving CTRL+C and CTRL+V encompasses a spectrum of behaviors varying in severity and intentionality. Analysis of institutional honor code violations and disciplinary records enables classification into distinct categories:
涉及 CTRL+C 和 CTRL+V 的当代学术不端行为涵盖了在严重性和故意性上各不相同的一系列行为。对机构荣誉准则违规和纪律记录的分析可以将其分类为不同的类别：
Direct Verbatim Copying / 直接逐字复制 The most unambiguous form of plagiarism involves unacknowledged reproduction of source text without quotation marks or citation. This traditional violation persists despite widespread availability of plagiarism detection software, suggesting

that some students either underestimate detection probabilities or lack understanding of citation requirements.
最明确的抄袭形式涉及在没有引号或引用的情况下未经承认地复制源文本。尽管抄袭检测软件广泛可用，这种传统违规行为仍然存在，表明一些学生要么低估了检测概率，要么缺乏对引用要求的理解。
Patchwork Plagiarism / 拼凑抄袭 Students may copy segments from multiple sources, assembling them into a coherent document with minimal original contribution. This "mosaic" approach creates the illusion of originality while constituting systematic intellectual appropriation. Detection of such violations often requires sophisticated textual analysis, as the resulting document may evade simple similarity detection algorithms.
学生可能从多个来源复制片段，将它们组装成一个具有最小原创贡献的连贯文档。这种"马赛克"方法创造了原创性的幻觉，同时构成了系统性的知识挪用。此类违规的检测通常需要复杂的文本分析，因为生成的文档可能规避简单的相似性检测算法。
AI-Generated Content Replication / AI 生成内容复制 The emergence of generative AI has introduced a novel category wherein students copy content produced by language models such as ChatGPT, Google Bard, or DeepSeek. This practice raises distinct ethical questions, as the copied material lacks traditional "authorship" yet represents intellectual output not originating from the student. Recent institutional policies have begun explicitly prohibiting unsubstantiated AI generation, with some universities implementing mandatory AI detection screening for thesis submissions.
生成式 AI 的出现引入了一个新类别，学生复制由 ChatGPT、Google Bard 或 DeepSeek 等语言模型产生的内容。这种做法提出了独特的伦理问题，因为复制的材料缺乏传统的"作者身份"， yet 代表了并非源自学生的智力产出。最近的机构政策已开始明确禁止未经证实的 AI 生成，一些大学对论文提交实施强制性 AI 检测筛查。
Data and Image Fabrication / 数据与图像伪造 Advanced violations involve using AI tools to generate or manipulate research data, experimental images, and statistical analyses. These practices extend beyond textual copying to encompass scientific fraud, as documented in recent investigations revealing students utilizing generative AI for "experimental image editing and substitution of autonomous design."
高级违规涉及使用 AI 工具生成或操纵研究数据、实验图像和统计分析。这些做法超越了文本复制，涵盖了科学欺诈，正如最近调查所记录的，学生利用生成式 AI 进行"实验图像编辑和自主设计替代"。

3.3 Disciplinary Variations / 学科差异

The prevalence and acceptance of copy-paste behaviors vary significantly across academic disciplines. Technical fields such as computer science and engineering may exhibit higher rates of code duplication, while humanities and social sciences demonstrate greater concern for textual plagiarism. However, the universal applicability of generative AI has somewhat homogenized these patterns, as AI tools can produce plausible content across disciplinary boundaries.
复制粘贴行为的普遍性和接受度在不同学科之间存在显著差异。计算机科学和工程等技术领域可能表现出更高的代码复制率，而人文和社会科学则表现出对文本抄袭的更大关注。然而，生成式 AI 的普遍适用性已在某种程度上使这些模式同质化，因为 AI 工具可以跨学科边界产生合理的内容。
Interdisciplinary research from Zhejiang University encompassing 20 institutions revealed that graduate students (master's and doctoral candidates) demonstrate higher rates of sophisticated copy-paste violations, including AI-assisted data fabrication, compared to undergraduates. This finding suggests that advanced students may exploit technological tools more effectively while simultaneously facing greater pressures to produce original research.
涵盖 20 所机构的浙江大学跨学科研究显示，与本科生相比，研究生（硕士和博士候选人）表现出更高比例的复杂复制粘贴违规行为，包括 AI 辅助的数据伪造。这一发现表明，高级学生可能更有效地利用技术工具，同时面临更大的生产原创研究的压力。

4. Institutional Responses and Detection Mechanisms / 机构应对与检测机制

4.1 Policy Frameworks and Honor Codes / 政策框架与荣誉准则

Universities worldwide have responded to the copy-paste epidemic through comprehensive policy revisions and technological implementations. In China, the Ministry of Education has mandated stringent academic integrity protocols, with institutions such as Fudan University promulgating explicit regulations regarding AI tool usage in undergraduate thesis preparation.
全球大学已通过全面的政策修订和技术实施应对复制粘贴流行。在中国，教育部已强制要求严格的学术诚信协议，复旦大学等机构已颁布关于本科论文准备中 AI 工具使用的明确规定。
Fudan University's *Regulations on the Use of AI Tools in Undergraduate Graduation Theses (Trial)* establishes six prohibited domains: research design, data analysis, raw data collection, results imaging, thesis writing, defense and inspection, and confidential content. Violations result in thesis failure and degree disqualification. This policy exemplifies the shift from reactive plagiarism detection to proactive AI usage governance.
复旦大学《关于在本科毕业论文（设计）中使用 AI 工具的规定（试行）》确立了六个禁止领域：研究设计、数据分析、原始数据收集、结果图像、论文撰写、答辩与检查以及涉密内容。违规导致论文不合格和学位取消。这一政策体现了从被动抄袭检测到主动 AI 使用治理的转变。
Similarly, Southwest University and Renmin University of China have implemented AIGC (AI-Generated Content) detection for 2025 undergraduate theses, with results serving as critical references for grade determination and excellence evaluation. These measures reflect institutional recognition that traditional plagiarism detection insufficiently addresses AI-generated content replication.
同样，西南大学和中国人民大学已对 2025 年本科毕业论文实施 AIGC（AI 生成内容）检测，结果作为成绩评定和评优的重要参考。这些措施反映了机构认识到传统抄袭检测不足以应对 AI 生成内容复制。

4.2 Technological Detection Systems / 技术检测系统

The arms race between duplication techniques and detection methodologies has intensified significantly. Contemporary detection ecosystems comprise multiple layers:
复制技术与检测方法之间的军备竞赛已显著 intensified。当代检测生态系统包括多个层次：
Textual Similarity Detection / 文本相似性检测 Platforms such as Turnitin, iThenticate, and CNKI (China National Knowledge Infrastructure) compare submissions against extensive databases of published literature, internet sources, and previously submitted student papers. These systems generate similarity indices that flag potential plagiarism for instructor review. However, their effectiveness against AI-generated content remains limited, as LLM outputs typically exhibit low similarity to existing sources.
Turnitin、iThenticate 和 CNKI（中国知网）等平台将提交内容与大量已发表文献、互联网资源和先前提交的学生论文数据库进行比较。这些系统生成相似性指数，标记潜在抄袭供教师审查。然而，它们对 AI 生成内容的有效性仍然有限，因为 LLM 输出通常与现有来源表现出低相似性。
AI Content Detection / AI 内容检测 Specialized tools designed to identify machine-generated text have emerged, including GPTZero, Originality.ai, and institutional implementations such as those deployed at Beijing University of Posts and Telecommunications and Communication University of China. These analyzers examine linguistic patterns, perplexity scores, and burstiness metrics to distinguish human from AI authorship. Nonetheless, reliability concerns persist, with studies indicating significant false positive and false negative rates.
专门设计用于识别机器生成文本的工具已经出现，包括 GPTZero、Originality.ai 以及北京邮电大学和中国传媒大学等部署的机构实施方案。这些分析器检查语言模式、困惑度分数和突发性指标，以区分人类与 AI 作者。然而，可靠性问题仍然存在，研究表明存在显著的假阳性和假阴性率。
Proctoring and Monitoring Software / 监考与监控软件 Examination contexts utilize proctoring tools such as Proctorio and Respondus LockDown Browser to restrict clipboard functionality and monitor keystroke patterns. A 2024 *Educause Review* study found that 60% of Canvas-based examinations employ proctoring software, with features including clipboard restrictions, screen recording, and keystroke logging that can flag suspicious paste actions.
考试环境利用 Proctorio 和 Respondus LockDown Browser 等监考工具限制剪贴板功能并监控按键模式。2024 年《Educause 评论》研究发现，60%的基于 Canvas 的考试使用监考软件，具有包括剪贴板限制、屏幕录制和按键记录等功能，可以标记可疑的粘贴行为。

4.3 Pedagogical Interventions / 教学干预

Beyond punitive and technological measures, institutions increasingly emphasize pedagogical strategies to reduce copy-paste reliance. Reflective writing instruction has demonstrated efficacy, with one study documenting a 33.13% reduction in plagiarism following implementation of reflective writing curricula. Key components include:
除了惩罚和技术措施外，机构越来越强调教学策略以减少复制粘贴依赖。反思性写作教学已证明有效，一项研究记录了实施反思性写作课程后抄袭减少 33.13%。关键组成部分包括：
Process-oriented assessment: Evaluating students on drafting stages, research logs, and revision histories rather than solely on final products, thereby making copy-paste submission more difficult.
过程导向评估：根据起草阶段、研究日志和修订历史评估学生，而非仅根据最终产品，从而使复制粘贴提交更加困难。
Citation skill instruction: Explicit teaching of paraphrasing techniques, summarization strategies, and disciplinary citation conventions to address knowledge deficits underlying unintentional plagiarism.
引用技能教学：明确教授改写技巧、总结策略和学科引用惯例，以解决导致无意抄袭的知识缺陷。
Authentic assignment design: Creating unique, context-specific prompts that resist generic or AI-generated responses, requiring integration of course-specific materials and personal analysis.
真实作业设计：创建独特、特定情境的提示，抵制通用或 AI 生成的回应，要求整合课程特定材料和个人分析。

5. Psychological and Educational Dimensions / 心理与教育维度

5.1 Motivational Factor**s / 动机因素**

Understanding why students resort to CTRL+C and CTRL+V requires analysis of motivational structures. Research identifies multiple contributing factors:
理解学生为何诉诸 CTRL+C 和 CTRL+V 需要分析动机结构。研究确定了多个促成因素：
Time Pressure and Workload / 时间压力与工作量 Contemporary university students face competing demands from coursework, employment, and extracurricular commitments. The efficiency of copy-paste methods presents an attractive solution to time scarcity, particularly when multiple assignments converge simultaneously. Survey data indicate that students perceive copy-paste behaviors as necessary adaptations to unrealistic academic expectations rather than moral failures.
当代大学生面临来自课程作业、就业和课外活动的竞争性需求。复制粘贴方法的效率为时间稀缺提供了有吸引力的解决方案，特别是当多个作业同时汇聚时。调查数据表明，学生将复制粘贴行为视为对不切实际学术期望的必要适应，而非道德失败。
Skill Deficits and Anxiety / 技能缺陷与焦虑 Many students lack confidence in their academic writing abilities, experiencing anxiety when confronted with complex synthesis tasks. The "fear of the blank page" drives reliance on existing textual resources as scaffolding. Notably, 88.0% of surveyed students in a 2019 China Youth Daily investigation believed universities should offer thesis writing courses, suggesting widespread recognition of skill deficiencies underlying copy-paste behaviors.
许多学生对自己的学术写作能力缺乏信心，在面对复杂的综合任务时经历焦虑。"空白页恐惧"驱使他们依赖现有文本资源作为脚手架。值得注意的是，2019 年《中国青年报》调查中 88.0%的受访学生认为大学应该开设论文写作课程，表明对复制粘贴行为背后技能缺陷的广泛认识。

Instrumental Orientation / 工具导向 Some students adopt purely instrumental approaches to education, viewing assignments as obstacles to credential acquisition rather than learning opportunities. Within this framework, copy-paste strategies represent rational optimization of effort-to-grade ratios, particularly in courses perceived as unrelated to career objectives.
一些学生对教育采取纯粹工具导向，将作业视为获取证书的障碍而非学习机会。在这一框架内，复制粘贴策略代表了努力与成绩比例的理性优化，特别是在被认为与职业目标无关的课程中。

5.2 Developmental Implications / 发展性影响

The developmental consequences of habitual copy-paste reliance extend beyond immediate academic integrity concerns. Longitudinal research suggests that students who routinely circumvent composition challenges through duplication fail to develop critical competencies including:
习惯性依赖复制粘贴的发展性后果超越了直接的学术诚信关注。纵向研究表明， routine 通过复制规避写作挑战的学生未能发展关键能力，包括：
Critical evaluation skills: The ability to assess source credibility, identify methodological limitations, and synthesize conflicting perspectives requires active engagement that copy-paste bypasses.
批判性评估技能：评估来源可信度、识别方法论局限性和综合冲突观点的能力需要复制粘贴绕过的主动参与。
Metacognitive awareness: Understanding one's own learning processes, knowledge gaps, and cognitive strategies—essential for lifelong learning—is undermined when students outsource intellectual labor to digital tools.
元认知意识：理解自己的学习过程、知识差距和认知策略——对终身学习至关重要——当学生将智力劳动外包给数字工具时受到削弱。
Ethical reasoning: Regular engagement in undisclosed copying may erode ethical sensitivity regarding intellectual property, establishing patterns of instrumental deception that transfer to professional contexts.
伦理推理：定期参与未披露的复制可能侵蚀关于知识产权的伦理敏感性，建立转移到职业环境的工具性欺骗模式。

5.3 **Cult**ural and Generational Factors / 文化与代际因素

Digital nativity shapes students' relationships with information and authorship. Students who have matured with internet technologies often exhibit different conceptualizations of intellectual property than previous generations. The "remix culture" prevalent in social media, where sharing and repurposing content is normalized, may conflict with academic expectations of original attribution.
数字原生性塑造了学生与信息和作者身份的关系。随着互联网技术成熟的学生通常表现出与前几代不同的知识产权概念。社交媒体中盛行的"混搭文化"，其中分享和重新利用内容被正常化，可能与学术对原始归属的期望冲突。
International students face additional challenges, as academic integrity norms vary across educational systems. What constitutes acceptable paraphrasing or common knowledge differs culturally, and students educated in systems emphasizing rote memorization may struggle with expectations of original synthesis unfamiliar in their prior educational experiences.
国际学生面临额外挑战，因为学术诚信规范因教育系统而异。什么构成可接受的改写或常识因文化而异，在强调死记硬背的教育系统中受教育的学生可能难以应对在其先前教育经历中不熟悉的原创综合期望。

6. Case Studies and Empirical Evidence / 案例研究与实证证据

6.1 Institutional Case: Zero-Tolerance Policies / 机构案例：零容忍政策

Recent media reports document a prominent case where a university instructor assigned zero grades to all AI-generated compositions in a first-year English course. The instructor's announcement in the course group chat stated that essays produced through AI tools would receive zero marks, reflecting institutional commitment to maintaining authentic assessment. Student interviews revealed surprise at detection capabilities, with one respondent noting: "I didn't expect the teacher would use AI detection tools. Honestly, AI use is quite common among my classmates, and I find it quite tempting."
最近的媒体报道记录了一个突出案例，某大学教师在大一英语课程中将所有 AI 生成的作文判为零分。教师在课程群聊中的公告声明，通过 AI 工具产生的论文将获得零分，反映了机构对维持真实评估的承诺。学生访谈揭示了对检测能力的惊讶，一位受访者指出："我没想到老师会用 AI 检测工具。说实话，我身边的同学对 AI 的使用确实挺普遍，而且我觉得诱惑性挺大的。"
This case illustrates both the effectiveness of explicit policy communication and the prevalence of copy-paste behaviors among students who may underestimate detection probabilities or overestimate AI output sophistication.
这一案例说明了明确政策沟通的有效性，以及可能低估检测概率或高估 AI 输出复杂性的学生中复制粘贴行为的普遍性。

6.2 Comparative Analysis: Pre-AI vs. Post-AI Eras / 比较分析：AI 前时代与 AI 后时代

Comparative data from the early 2000s and 2024-2025 reveal exponential growth in duplication-related misconduct. Research from the University of Florida documented that internet-based "copy-paste" plagiarism increased fourfold between 1999 and 2010. Current data suggest that generative AI has precipitated a comparable magnitude of increase in the 2020s, with the added complexity that AI-generated content often evades traditional similarity detection.
2000 年代初与 2024-2025 年的比较数据揭示了与复制相关的不端行为的指数增长。佛罗里达大学的研究记录，基于互联网的"复制粘贴"抄袭在 1999 年至 2010 年间增加了四倍。当前数据表明，生成式 AI 在 2020 年代引发了类似幅度的增长，增加了复杂性，因为 AI 生成的内容通常规避传统相似性检测。
The transition from "copying from the internet" to "copying from AI" represents not merely a quantitative escalation but a qualitative shift in academic misconduct. While internet plagiarism involved appropriating human-created con-

tent, AI replication involves presenting machine-generated text as human intellectual product, raising distinct questions about authorship, originality, and the nature of academic work.
从"从互联网复制"到"从 AI 复制"的转变不仅代表了学术不端的数量升级，而且是质性转变。虽然互联网抄袭涉及挪用人类创作的内容，但 AI 复制涉及将机器生成的文本呈现为人类智力产品，提出了关于作者身份、原创性和学术工作性质的不同问题。

6.3 Longitudinal Student Trajectories / 学生纵向轨迹

Qualitative research tracking individual students across their academic careers reveals developmental patterns in copy-paste behaviors. Many students report "experimenting" with copying during early undergraduate years, particularly in large lecture courses with high enrollment and standardized assignments. However, students who receive explicit instruction in research methodologies and academic writing often transition toward more sophisticated, ethical engagement with sources.
追踪学生整个学术生涯的定性研究揭示了复制粘贴行为的发展模式。许多学生报告在本科早期"尝试"复制，特别是在招生规模大、作业标准化的大班讲座课程中。然而，接受研究方法和学术写作明确指导的学生通常转向与来源更复杂、更道德的接触。
Conversely, students who experience early success through undisclosed copying without detection or consequence may develop habitual patterns that persist into advanced coursework and thesis preparation. These trajectories underscore the importance of early intervention and consistent enforcement of academic integrity standards.
相反，早期通过未披露复制而未被发现或产生后果的学生可能发展成持续到高级课程和论文准备的习惯模式。这些轨迹强调了早期干预和一致执行学术诚信标准的重要性。

7. Ethical Frameworks and Philosophical Considerations / 伦理框架与哲学考量

7.1 Redefining Authorship in the Digital Age / 数字时代作者身份的重新定义

The proliferation of AI-generated content necessitates philosophical examination of authorship concepts. Traditional academic frameworks assume clear boundaries between original creation and appropriation, with human intentionality serving as the distinguishing criterion. However, human-AI collaborative writing blurs these boundaries, raising questions about the threshold at which AI assistance becomes AI substitution.
AI 生成内容的泛滥 necessitates 对作者身份概念的哲学审视。传统学术框架假设原创与挪用之间有明确界限，人类意向性作为区分标准。然而，人机协作写作模糊了这些界限，提出了关于 AI 辅助何时成为 AI 替代的阈值问题。
Institutional policies increasingly distinguish between "AI-assisted" writing—where tools support research, grammar checking, or idea generation—and "AI-generated" writing where the machine produces substantive content subsequently copied by the student. This distinction attempts to preserve the developmental value of academic writing while acknowledging legitimate technological assistance.
机构政策越来越区分"AI 辅助"写作——工具支持研究、语法检查或想法生成——和"AI 生成"写作——机器产生随后被学生复制的实质性内容。这种区分试图在承认合法技术辅助的同时，保留学术写作的发展价值。

7.2 Information Commons vs. Intellectual Property / 信息公地 vs. 知识产权

The copy-paste debate reflects broader tensions between open information sharing and proprietary intellectual property rights. Proponents of open access and knowledge democratization argue that strict plagiarism enforcement perpetuates gatekeeping mechanisms that disadvantage students from under-resourced educational backgrounds. Conversely, intellectual property advocates maintain that attribution requirements protect marginalized creators and ensure proper credit for intellectual labor.
复制粘贴辩论反映了开放信息共享与专有知识产权之间更广泛的紧张关系。开放获取和知识民主化的支持者认为，严格的抄袭执法 perpetuates 使资源不足教育背景学生处于不利地位的守门机制。相反，知识产权倡导者坚持认为，归属要求保护边缘化创作者并确保对智力劳动的适当认可。
A mediated position recognizes that academic integrity policies serve not merely to protect property rights but to ensure the authenticity of credentialing. When students submit copied work, they misrepresent their own knowledge and capabilities, undermining the informational value of academic degrees for employers and society.
一种调和立场认识到，学术诚信政策不仅服务于保护产权，而且确保认证的真实性。当学生提交复制的作品时，他们 misrepresent 自己的知识和能力，削弱学术学位对雇主和社会的信息价值。

7.3 Pedagogical vs. Punitive Approaches / 教学性 vs. 惩罚性方法

Ethical frameworks for addressing copy-paste behaviors range from punitive to pedagogical orientations. Punitive approaches emphasize detection, sanction, and deterrence through severe penalties including course failure, suspension, or expulsion. While effective for egregious violations, purely punitive frameworks may fail to address underlying skill deficits or motivational factors driving misconduct.
解决复制粘贴行为的伦理框架范围从惩罚性到教学性导向。惩罚性方法强调通过严厉处罚包括课程不及格、停学或开除进行检测、制裁和威慑。虽然对严重违规有效，但纯粹惩罚性框架可能无法解决导致不端行为的潜在技能缺陷或动机因素。
Pedagogical approaches prioritize education, skill development, and assignment redesign to reduce opportunities and incentives for copying. Research supports the efficacy of these interventions: institutions implementing comprehensive writing instruction and process-oriented assessment report significant reductions in plagiarism rates compared to those relying solely on detection software.
教学方法优先考虑教育、技能发展和作业重新设计，以减少复制的机会和激励。研究支持这些干预的有效性：实施综合写作指导和过程导向评估的机构报告，与仅依赖检测软件的大学相比，抄袭率显著降低。
Integrated frameworks combine proportionate sanctions for violations with robust support for academic skill devel-

opment, recognizing that effective integrity promotion requires both consequences for misconduct and elimination of its root causes.
综合框架结合对违规的比例制裁与对学术技能发展的强有力支持，认识到有效的诚信促进需要对不端行为的后果和消除其根源。

8. Future Directions and Recommendations / 未来方向与建议

8.1 Technological Evolution / 技术演进

The copy-paste landscape will continue evolving as AI technologies advance. Next-generation language models will produce increasingly sophisticated text indistinguishable from human writing, challenging detection methodologies. Concurrently, AI-assisted research tools will become more deeply integrated into academic workflows, necessitating continual policy adaptation.
随着 AI 技术进步，复制粘贴格局将继续演变。下一代语言模型将产生越来越复杂的文本，与人类写作无法区分，挑战检测方法。同时，AI 辅助研究工具将更深入地整合到学术工作流程中， necessitating 持续的政策适应。
Emerging technologies such as blockchain-based attribution systems and real-time writing analytics may provide new mechanisms for verifying authorship and documenting writing processes. Investment in these technologies should accompany continued refinement of detection capabilities.
区块链归属系统和实时写作分析等新兴技术可能提供验证作者身份和记录写作过程的新机制。对这些技术的投资应伴随检测能力的持续改进。

8.2 Policy Recommendations / 政策建议

Based on current evidence, we recommend the following policy directions:
基于当前证据，我们建议以下政策方向：
Explicit AI Usage Policies: Institutions should develop clear, specific guidelines distinguishing permissible from prohibited AI applications, with explicit mention of copy-paste behaviors involving machine-generated content.
明确的 AI 使用政策：机构应制定清晰、具体的指导方针，区分可允许和禁止的 AI 应用，明确提及涉及机器生成内容的复制粘贴行为。
Integrated Detection Ecosystems: Rather than relying on single detection tools, universities should implement multi-layered verification systems combining similarity detection, AI-content analysis, and process-oriented assessment data.
综合检测生态系统：大学不应依赖单一检测工具，而应实施结合相似性检测、AI 内容分析和过程导向评估数据的多层验证系统。
Mandatory Academic Literacy Instruction: All students should complete coursework in research ethics, citation practices, and effective paraphrasing before undertaking thesis work, addressing the skill deficits underlying many copy-paste violations.
强制性学术素养教学：所有学生在进行论文工作前应完成研究伦理、引用实践和有效改写课程，解决许多复制粘贴违规背后的技能缺陷。
Faculty Development: Instructors require training in AI tool capabilities and limitations, detection technology usage, and assignment design strategies that promote authentic engagement.
教师发展：教师需要接受 AI 工具能力和局限性、检测技术使用以及促进真实参与作业设计策略的培训。

8.3 Pedagogical Innovation / 教学创新

Educational responses to copy-paste challenges should emphasize:
对复制粘贴挑战的教育回应应强调：
Authentic Assessment: Designing evaluations that require application of knowledge to novel situations, integration of personal experience, or demonstration of skills in controlled environments where copying is impractical.
真实评估：设计评估，要求将知识应用于新情境、整合个人经验，或在复制不切实际的受控环境中展示技能。
Process Documentation: Requiring submission of research notes, draft iterations, and reflective commentaries alongside final papers to ensure engagement with the compositional process.
过程记录：要求提交研究笔记、草稿迭代和反思评论以及最终论文，以确保参与写作过程。
Collaborative Writing: Implementing peer review and collaborative composition assignments that develop students' abilities to provide and incorporate feedback, reducing reliance on external sources.
协作写作：实施同伴评审和协作写作作业，培养学生提供和整合反馈的能力，减少对外部来源的依赖。
Critical AI Literacy: Teaching students to evaluate AI-generated content critically, recognizing its limitations and potential biases, rather than prohibiting or uncritically accepting such tools.
批判性 AI 素养：教学生批判性评估 AI 生成内容，认识其局限性和潜在偏见，而非禁止或盲目接受此类工具。

9. Conclusion / 结论

The keyboard shortcuts CTRL+C and CTRL+V represent more than mere technological conveniences; they embody the fundamental tensions of digital age education between efficiency and authenticity, access and originality, assistance and substitution. Our analysis reveals that these tools, while offering legitimate benefits for information management and research synthesis, have facilitated widespread academic misconduct ranging from unintentional citation errors to systematic AI-generated content replication.
键盘快捷键 CTRL+C 和 CTRL+V 代表的不仅仅是技术便利；它们体现了数字时代教育在效率与真实性、获取与原创性、辅助与替代之间的根本张力。我们的分析表明，这些工具虽然为信息管理和研究综合提供了合法益处，但已促成了从无意引用错误到系统性 AI 生成内容复制的广泛学术不端。
Current data from Chinese higher education institutions indicate that nearly 30% of students utilize generative AI primarily for assignment completion, with significant proportions engaging in direct copy-paste of machine-generated content. Institutional responses have evolved from traditional plagiarism detection to comprehensive AI governance frameworks, exemplified by Fudan University's prohibition of AI usage in critical research components and the implementation of AIGC detection at multiple universities.

来自中国高等教育机构的当前数据表明，近 30%的学生主要利用生成式 AI 完成作业，相当比例的学生从事机器生成内容的直接复制粘贴。机构应对措施已从传统抄袭检测演变为全面的 AI 治理框架，以复旦大学禁止在关键研究环节使用 AI 以及多所大学实施 AIGC 检测为典范。

Effectively addressing these challenges requires integrated approaches combining technological detection, policy enforcement, and pedagogical innovation. Punitive measures alone prove insufficient; institutions must simultaneously address the skill deficits, time pressures, and motivational factors driving students toward copy-paste shortcuts. The 33.13% reduction in plagiarism observed following reflective writing instruction demonstrates that educational interventions can significantly impact student behaviors.

有效应对这些挑战需要结合技术检测、政策执行和教学创新的综合方法。单靠惩罚措施是不够的；机构必须同时解决促使学生走向复制粘贴捷径的技能缺陷、时间压力和动机因素。反思性写作教学后观察到的 33.13%抄袭减少表明，教育干预可以显著影响学生行为。

Ultimately, the goal is not to eliminate CTRL+C and CTRL+V from academic workflows—these tools remain valuable for legitimate research activities—but to cultivate student capacities for critical evaluation, ethical reasoning, and original synthesis that render mere duplication intellectually unsatisfying and professionally unnecessary. As generative AI capabilities advance, the distinction between human and machine authorship will increasingly blur, making the development of authentic academic voice and rigorous analytical skills more essential than ever.

最终目标不是从学术工作流程中消除 CTRL+C 和 CTRL+V——这些工具对合法研究活动仍然有价值——而是培养学生批判性评估、伦理推理和原创综合的能力，使 mere 复制在智力上不能令人满意且在职业上不必要。随着生成式 AI 能力的进步，人类与机器作者身份之间的区别将 increasingly 模糊，使真实学术声音和严格分析技能的发展比以往任何时候都更加 essential。

The future of academic integrity in the digital age depends not on technological prohibition but on educational transformation—fostering environments where original thought is nurtured, authentic voice is valued, and the shortcuts offered by CTRL+C and CTRL+V are recognized as inadequate substitutes for genuine intellectual engagement.

数字时代学术诚信的未来不取决于技术禁止，而取决于教育转型——培养滋养原创思想、重视真实声音的环境，并认识到 CTRL+C 和 CTRL+V 提供的捷径无法替代真正的智力参与。

致谢

谢 SHIT

Data Availability / 数据可用性

The data supporting this study are derived from publicly available institutional reports, published surveys by MyCos Research (麦可思研究), and academic publications cited in the references. Specific datasets analyzed during the current study are available from the corresponding author upon reasonable request.

支持本研究的数据来自公开的机构报告、麦可思研究发表的调查和参考文献中引用的学术出版物。当前研究期间分析的具体数据集可根据合理要求从通讯作者处获取。

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